

The Utility Problem

Amidst the rising occurrence of catastrophic wildfires, the significance of wildfire risk assessment comes to the forefront as an essential instrument for protecting communities, infrastructure, and ecosystems. By finding areas that are at risk ahead of time, this evaluation can help start smart planning, quick actions, and working together to lessen the big effects of these fires on people, things, and the environment.

The objective lies in the assessment and mitigation of wildfire risk concerning electricity transmission and distribution utility assets, striving to comprehend the assets' susceptibility to the looming wildfire hazard.

The assessment is usually handled from two points of view:

1. Risk of utility assets (like powerlines) causing wildfire - addressing the risk of transmission or distribution utility assets, like powerlines, causing wildfires is important. An analytical system helps identify high-risk areas, manage vegetation, and deploy monitoring solutions. This ensures power distribution reliability while minimizing the threat of asset-related wildfires to the environment and communities.
2. Risk of utility assets being impacted by wildfire - evaluating the risk of transmission or distribution utility assets being impacted by wildfires is vital. An analytical system identifies vulnerable areas, strengthens infrastructure, and enhances emergency response, minimizing disruptions and ensuring community safety. The backup or redundant options might be proposed to increase the robustness of the system and prevent unnecessary outages.

Overall, an analytical system for versatile wildfire risk analysis is vital as it equips electric utilities with data-driven insights crucial for safeguarding infrastructure and communities against potential wildfire events. By integrating diverse data sources, it enables utilities to swiftly identify high-risk areas, prioritize actions, customize mitigation strategies, ensure operational continuity, and enhance communication with stakeholders, thus fostering resilience, regulatory compliance, and public trust in the face of evolving environmental challenges.

The Data Solutions and Methods

For a successful Wildfire Risk Assessment analysis, multiple data sources are required. These data sources are:

- Geospatial Information System (GIS) data - important pieces of information that these data carry is the geolocation of assets.
- Asset Information Data - includes important attributes of the grid assets like transformer size, power conductor length or insulation, etc.

- Vegetation survey and maps - many times the utilities do such surveying by themselves or public/private data sources can be obtained from the internet.
- Weather information - for usual weather patterns.

The Awesense Energy Transition Platform supports the majority of the above-mentioned data sources. Once the data are ingested into the platform, the Awesense Energy Data Model (EDM) APIs allow users to query these data and develop their own analysis based on detailed business needs.

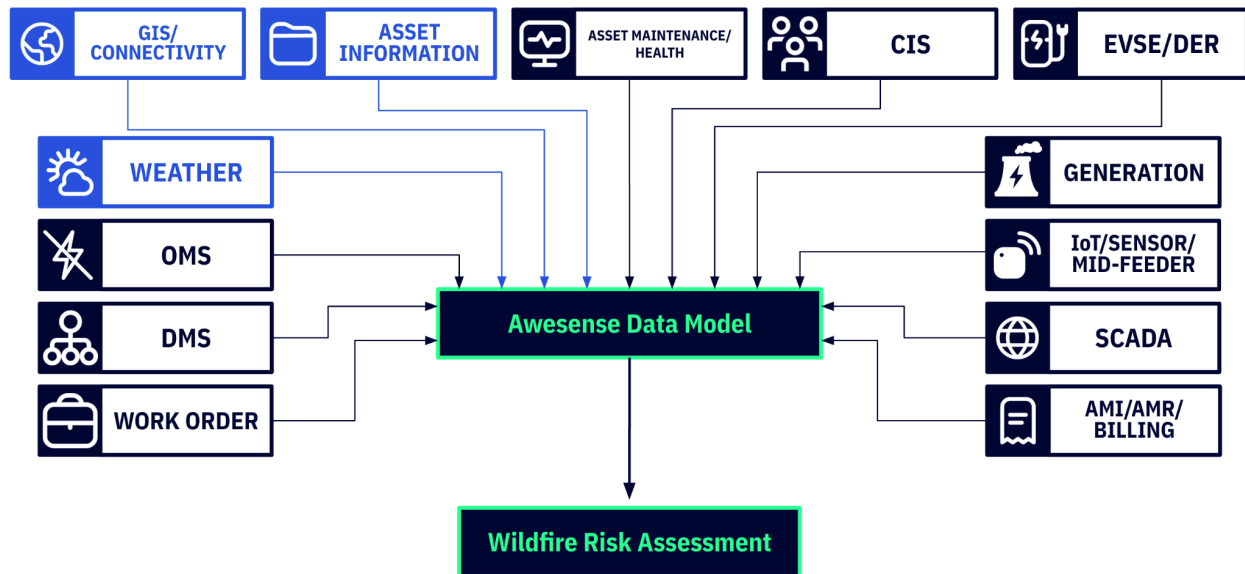
In terms of choosing the right methodology for such assessment, it should not matter from the point of view, if the risk to the environment or risk to assets should be assessed. The high-level methodology for Wildfire Risk Assessment is as follows:

- Identify High-Risk Areas - determine which areas have a higher probability of experiencing wildfires based on factors such as vegetation density, historical fire occurrence, and weather patterns.
- Assess Asset Vulnerability - evaluate how susceptible your electric utility assets are to fire damage. This assessment may consider factors like asset type, age, condition, and criticality.
- Calculation of Overall Risk - combine the information from the fire risk assessment and asset vulnerability assessment to calculate an overall risk score for each asset or area.
- Prioritize Mitigation Efforts - with the risk assessment results, prioritize mitigation efforts based on the identified high-risk assets. Develop strategies to reduce the impact of wildfires on these assets, such as improving vegetation management, implementing fire-resistant designs, and developing emergency response plans. Based on this analysis, it is also possible, for example, to route the new line along a different, more suitable route in terms of fire safety. This will also reduce the risk of damage to other parts of the grid, such as transformers, etc.
- Enhance Preparedness and Resilience - through this assessment and mitigation process, you aim to enhance the preparedness and resilience of your electricity transmission and distribution utility against the potential threat of wildfires. This includes having a clear understanding of the risks, having plans in place to respond effectively, and taking proactive measures to minimize potential damage.

Awesense Platform is a versatile analytical tool that can be used as the foundation for the development of Wildfire Risk Assessment since its architecture and features allow for a combination of data sources with the aim of delivering the above-mentioned methodology steps.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. Please contact [Awesense](#) if you are interested in the implementation of this use case.